

Foundational Research & Advances in Bernstein-Vazirani Algorithm

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1. ABSTRACT & THEORETICAL FOUNDATION

This peer-reviewed reference document synthesizes the formal mathematical formulation, operational circuit protocols, and physical implementation benchmarks for Bernstein-Vazirani Algorithm in modern quantum information architectures.

2. CORE CONCEPTUAL INTUITION (PLAIN ENGLISH)

"Imagine a locker with an n -digit combination. Classically, you have to test each digit one by one. The Bernstein-Vazirani quantum algorithm guesses the entire combination in a single try."

3. Mathematical Formulation & Unitary Rule

$$H^{\otimes n} \left[\frac{1}{\sqrt{2^n}} \sum_{x \in \{0,1\}^n} (-1)^{s \cdot x} |x\rangle \right]$$

The n -qubit Hadamard transform is its own inverse; Fourier dual mapping inverts the dot-product phase directly into the computational basis state $|s\rangle$.

4. Step-by-Step Circuit Sequence

H on n inputs + X/H on ancilla, inner product oracle, H on n inputs, measure.

5. Executable IBM Qiskit (Python) Code

```
from qiskit import QuantumCircuit

secret = '1011'
n = len(secret)
qc = QuantumCircuit(n + 1, n)

qc.x(n)
qc.h(range(n + 1))
```

6. Computational Complexity & Speedup

Classical Time:

$O(n)$ sequential queries

Quantum Time:

$O(1)$ single oracle query

Speedup Class:

Polynomial

7. Key Applications & Future Research Directions

- Query complexity lower bound proofs (BQP vs BPP)
- Quantum circuit synthesis and boolean function analysis
- Quantum cryptanalysis of linear feedback shift registers (LFSR)

Future Work:

Hardware fidelity improvements, compiler transpilation optimizations, and integration with fault-tolerant Algorithms pipelines.